

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

Paper 2 Core Paper

9648/02

September 2014

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
4	
5	
6	
7	
Section B	20
Total	100

This document consists of **20** printed pages and **no** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 Lactose intolerance in humans is the inability to hydrolyse lactose due to the lack of the enzyme lactase in the alimentary canal. As a result, bacteria in the large intestines feed on the lactose and produces fatty acids and methane which lead to diarrhoea and flatulence.

Bacteria have been used to produce lactase on an industrial scale as a dietary supplement for people who are lactose intolerant. Human lactase consists of a single 160 kDa polypeptide chain that localizes to the brush border membrane of intestinal epithelial cells.

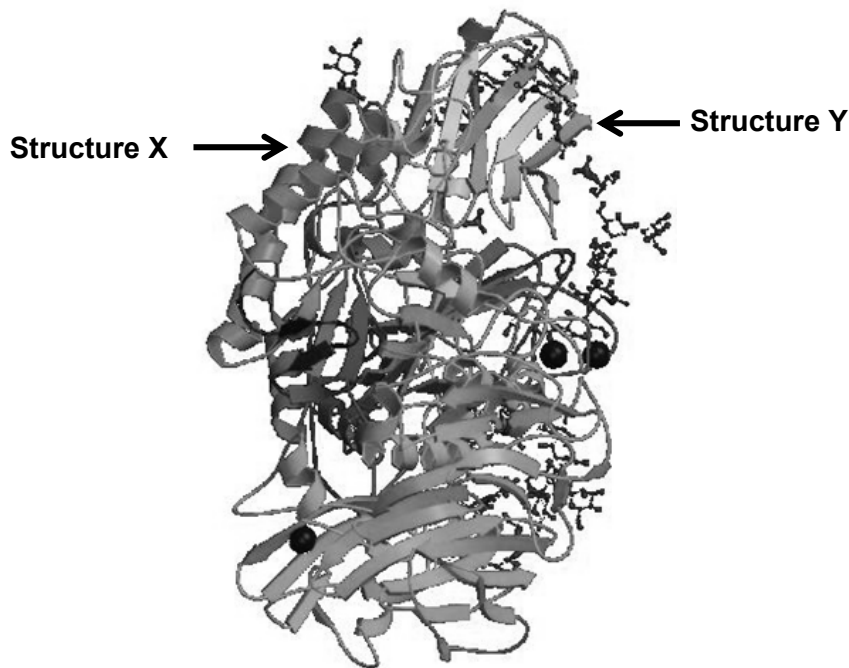


Fig. 1.1

- (a) With reference to the **Fig. 1.1**,
(i) state the levels of organization seen in the structure of lactase.

primary, secondary, tertiary structure;

[1]

- (ii) Describe structures **X** and **Y**

structure X- (alpha helix)

repeated coiling of polypeptide chain, intrachain hydrogen bonding between neighbouring N-H group and C=O group of peptide bonds four amino acids away;

Structure Y- (beta pleated sheets)

repeated folding of polypeptide chain, intrachain hydrogen bond between N-H group of a peptide bond on one fold and C=O of a peptide bond on another fold;

[2]

(b) Fig. 1.2 shows the hydrolysis of lactose.

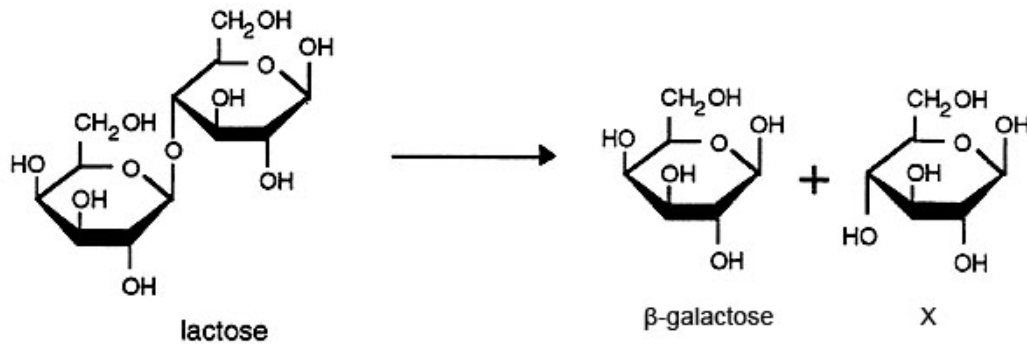


Fig. 1.2

(i) Describe the hydrolysis of lactose, naming the bond that is broken and product X.

Use of water to break β-1,4-glycosidic bond;
to form (either) galactose and β-glucose;

[2]

(c) Both lactase and lactose are protein and carbohydrate polymers respectively. Explain why there are fewer types of carbohydrate polymers compared to protein polymers.

lactose (carbohydrate) is made up of two monomers, glucose and galactose but lactase (proteins) are made up of many different types/ 20 types of monomers, amino acids;

amino acids differ by their R groups;

the different sequences and number of amino acids making up proteins give rise to the large number of different types of proteins / ref to various types of amino acids in a single polypeptide chain vs in a carbohydrate polymer, the same monomer makes up the polymer;

[2]

Cell organelles can be separated by centrifuging a cell extract in a sucrose density gradient. The organelles settle at the level in the sucrose solution which has the same density as their own.

The cells used to synthesize lactase were lysed and the cell extract centrifuged in a sucrose density gradient. Three distinct fractions of nuclei, mitochondria and ribosomes (in no particular order) were obtained. The three fractions **A**, **B** and **C** are shown in the Fig. 1.3.

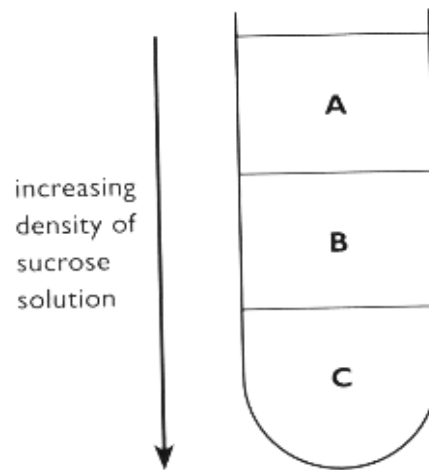


Fig. 1.3.

(d) Identify the organelles in each fraction and describe its role in the synthesis of lactase.

A: Ribosomes – translate genetic message carried by messenger RNA into a polypeptide chain;

B: Mitochondria – sites of cellular respiration, producing energy in the form of ATP which can then be used for peptide bond formation / amino acids activation / exocytosis;

C: Nucleus – contains nucleolus which is responsible for synthesis of ribosomal ribonucleic acids (rRNA) which is a component of ribosomes; OR contains genetic materials (DNA) and the genes found on DNA contains information on how lactase is synthesized;

1 mark for correct identification of all fractions and 1 mark each for each correct function]

[4]

[Total: 11]

- 2 The main function of a chromosome is to act as a **template** for the synthesis of RNA molecules, since only in this way does the genetic information stored in chromosomes become directly useful to the cell. RNA synthesis is a highly selective process. In most mammalian cells, only about 1% of the genetic information is transcribed into functional RNA sequences.

Before a cell can divide, it must produce a new copy of each of its chromosomes. Thus cell division is preceded by a special 'DNA-synthesis phase', during which each chromosome is replicated to produce two chromatids. These remain joined together at the centromere until the mitosis that soon follows.

- (a) Explain what is meant by the term *template* (line 1).

one strand of the double stranded DNA molecule;

where the nucleotide base sequence will determine the sequence of the mRNA that is transcribed from it;

sequence on DNA template is complementary, A-U, G-C, to the mRNA transcribed from it;

[3]

- (b) Suggest why only about 1% of genetic information is transcribed into functional RNA sequences in most mammalian cells.

due differential gene expression/ cell specialization/ differentiation, only some genes are transcribed and translated/ while other genes that encode proteins not required by the cell are silenced/ turned off;

presence of non-coding sequences such as centromeres/ telomeres/ introns, which are not found in functional RNA sequences;

[2]

- (c) Outline how messenger RNA is synthesized in a eukaryotic cell.

RNA polymerase recognises and binds to the double-stranded DNA at a promoter / TATA binding protein (TBP) binds to TATA box/sequence, thus allowing RNA polymerase to bind; unwinding the DNA; ® *Pribnow box*

elongation, nucleotides added covalently to the 3' -OH end of the growing RNA/ new RNA strand is formed from the 5' to 3' direction;

ref to complementary base pairing during the addition of nucleotides by RNA polymerase;

termination occurs after the polyadenylation signal sequence/ AAUAAA found on the DNA template strand, is transcribed;

[3]

The graphs in **Fig. 2.1** show the amount of DNA in the nuclei of cells taken from different parts of a mammalian testis during the formation of sperm cells. The formation of sperm cells involves mitosis and meiosis.

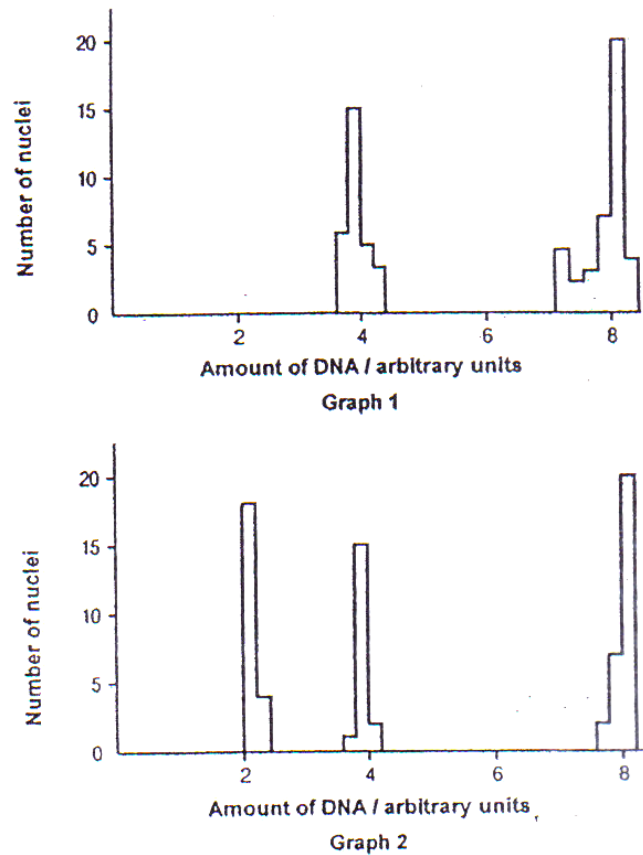


Fig. 2.1

(d) With reference to **Fig. 2.1**,

(i) explain the evidence which shows that Graph 1 represents cells undergoing mitosis.

Peak at about 4 a.u. represents cells at G1 phase / after telophase;

peak at about 8 a.u. represents cells at synthesis phase / interphase / after semi-conservative replication;

[2]

(ii) explain how the events of meiosis result in three groups of nuclei in Graph 2.

Separation of homologous chromosomes during Meiosis I / Anaphase I results in nuclei having DNA content at peak of about 4 a.u. (from peak of about 8 a.u.);

Separation of sister chromatids during Meiosis II / Anaphase II results in nuclei having DNA content at peak of about 2 a.u.;

- 3 The Ebola outbreak in West Africa is the world's deadliest to date and the World Health Organization (WHO) has now declared an international health emergency in response. It was first reported in February and according to the UN, 1013 people have died in Guinea, Liberia, Sierra Leone and Nigeria. (Why Ebola is so dangerous, 12 August 2014, BBC News)

Fig. 3.1 below shows some of the main structural features of the Ebola virus.

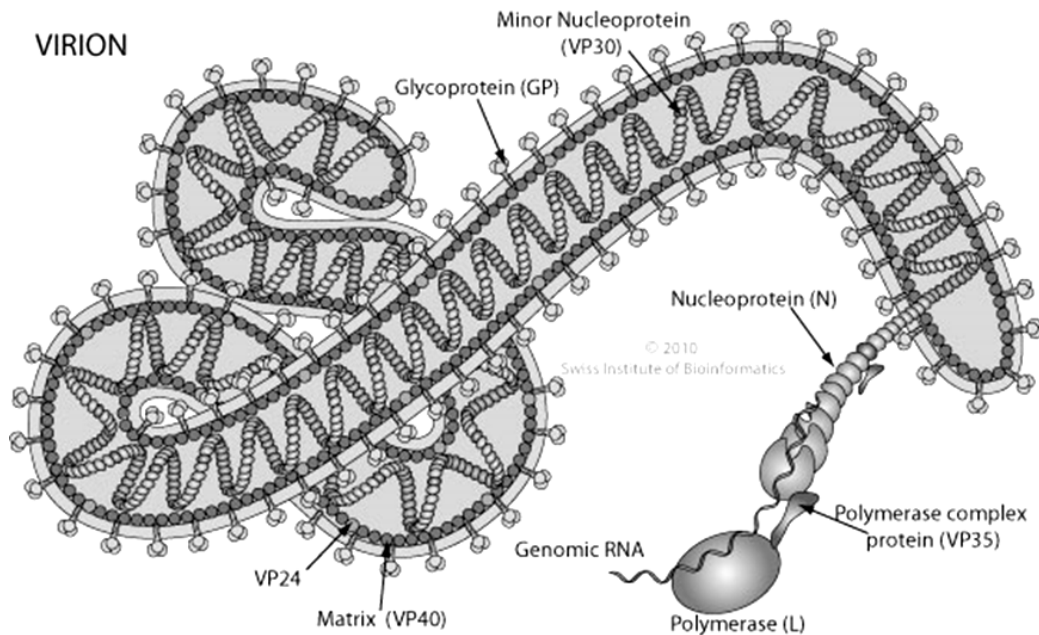


Fig. 3.1

The Ebola virus (EBOV) is similar to the influenza virus in many ways. EBOV infects the host cell in the same way as influenza does.

- (a) Using your knowledge of influenza, suggest how EBOV may infect a host cell.
GP binds to specific receptor molecules on host cell surface membrane;

virus enters via vesicle/ endosome through the invagination of the cell surface membrane by the process endocytosis;

Viral envelope fuses with endosome's membrane, exposing the capsid to digestion by cellular enzyme in the cytosol;

The genome of EBOV contains one molecule of linear, negative sense RNA which is 19kb long. The genome encodes seven proteins, which include: nucleoprotein (NP), VP35, VP40, glycoprotein (GP), VP30, VP24, and L protein, which provides RNA-dependent RNA activity (Leroy *et al.* 2001).

- (b) The EBOV genome encodes L protein. Explain why such a gene is needed by the virus.

L protein is an **RNA-dependent RNA polymerase**;

transcribes/ synthesises negative sense RNA into **positive strand RNA**/ uses the negative sense RNA as a template to synthesis a positive strand;

so that the positive sense RNA can be translated to form viral proteins;

Host cell lack RNA dependent RNA polymerase;

[3]

- (c) State how, despite having no ribosomes, EBOV is able to trigger the synthesis of its viral proteins.

uses host ribosomes;

[1]

An experimental drug known as ZMapp has been approved by WHO to be used on three Ebola patients.

- (d) Suggest how antiviral drugs can be used to halt the reproductive cycle of EBOV.

inhibitors; (compulsory point)

binds to proteins on virus, to prevent GP from binding to cell surface receptors thus preventing entry/ exit;

inhibits enzymes such as L protein, to prevent synthesis of positive sense RNA;

drugs such as ribozymes that will target viral RNA and cleaves them;

monoclonal antibodies;

antibodies bind to glycoproteins on surface of bacteria, targeting them for degradation by host immune system;

[2]

- (e) Even though Ebola is such a virulent disease, suggest why there is little progress in drug development.

lack of interest by pharmaceuticals;

research is expensive, and drug has to be given cheaply because disease is in 3rd world country;

more lucrative drugs to develop;

RNA virus, **no proof-reading in replication** makes development of vaccine difficult as **surface antigens change at a high rate**; OWTTE

[1]

- 4 Fig. 4.1 below shows the processes involved in gene expression in a mammalian cell.

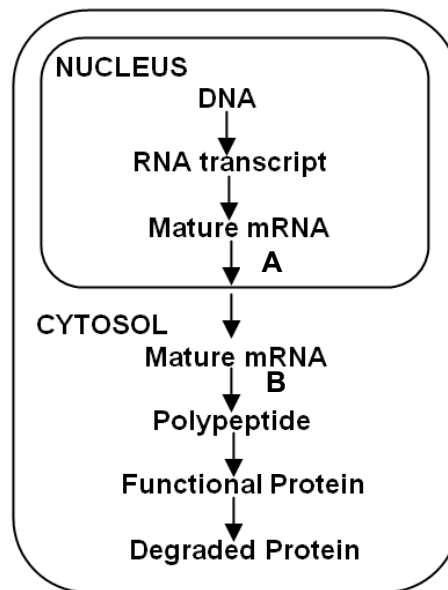


Fig. 4.1

- (a) (i) Regulation of gene expression at A occurs in eukaryotes. State why it is absent in prokaryotes.

Prokaryotes lack a nuclear envelope, no regulation at/by the membrane does not occur;

[1]

- (ii) Briefly describe one regulatory mechanism that can occur at process B.

Phosphorylation/ dephosphorylation to activate/inactivate translation initiation factors needed to initiate ribosomal binding;

Translational repressors bind at the 5' UTR preventing initiation or prevent ribosome from progressing;

Length of poly-A tail: The more stable an mRNA molecule is, the longer it is translated and more polypeptides are produced;

Antisense RNA that bind to the mRNA to prevent translation: SiRNA binds to mRNA forming a duplex RNA, preventing the ribosome from binding to translate mRNA, mRNA degrades;

Localisation of mRNA to concentrate the mRNA content in a specific location within the cell;

Cytoplasmic polyadenylation where cytoplasmic polymerase catalyse the elongation of the polyA tail;

[1]

The Hayflick Limit is the number of times a normal human cell population will divide until cell division stops. Evidences have shown that telomeres associated with each cell's DNA will get slightly shorter with each new cell division until they shorten to a critical length. This is due to the end-replication problem that eukaryotic chromosomes have.

- (b) Outline the end-replication problem in eukaryotic chromosomes.

DNA polymerase can only add nucleotides to the 3'OH end of a elongating / pre-existing polynucleotide;

When primer (bound to the very end of the template strand) removed but cannot be

replaced with DNA / gap cannot be filled in (because no 3'-OH available);

5' end of the DNA becomes shorter relative to that of the previous generation/
occurrence of a 3' overhang;

[3]

- (c) Describe the role of telomeres in an eukaryotic cell.

Ends of a linear eukaryotic chromosome which consists of short tandem repeats /
3'TTAGGG5' of a DNA sequence;

Protects genes from being eroded as DNA shortens due to the end-replication problem;

prevents the ends of the telomere from being degraded by exonuclease;

prevents the ends of the chromosomes from accidentally fusing with each other;

counting mechanism for the number of cell divisions that the cell has undergone;

[2]

- (d) Telomerase activity is essential in single celled organisms, while in humans and other vertebrates, it is only active in reproductive cells. Explain why regeneration of telomeres is not normally required in human somatic cells.

cells only required to divide a limited number of times / ref. to apoptosis/programmed cell death

[1]

- (e) Telomerase activity is generally reactivated in tumour cells. Explain how this knowledge could be employed in the fight against cancer.

Telomerase activity allow cancer cells to **divide indefinitely** because DNA does not shorten with each replication;

If telomerase inhibitors are used to hinder telomerase activity / stop regeneration of telomeres, this will limit number of division and triggers apoptosis

OR

shut down expression of telomerase gene so as to stop production of telomerase

[2]

[Total: 10]

- 5 The coat colours of Labrador Retriever dogs are determined by two pairs of unlinked genes B/b and E/e. The genotype of black coated dogs is B-E-, brown bbE- and golden --ee (where – indicates the presence of either the dominant or recessive allele).

(a) If heterozygous black (BbEe) dogs are mated, state the phenotypic ratio that can be expected in the progeny.

Show how you derive your answer.

Parental generation

Parental phenotypes

Parental genotypes

Black

BbEe

x

Black

BbEe

;

After meiosis,

gametes

produced

By random fertilization,

BE

bE

Be

be

BE

bE

Be

be

;

	BE	bE	Be	be
BE	BBEE Black	BbEE Black	BBEe Black	BbEe Black
bE	BbEE Black	bbEE Brown	BbEe Black	bbEe Brown
Be	BBEe Black	BbEe Black	BBee Golden	Bbee Golden
be	BbEe Black	bbEe Brown	Bbee Golden	bbee Golden

F₁ genotypic ratio

1 BBEE, 2 BbEE,
2BBEe, 4BbEe

1 bbEE, 2 bbEe

1 BBee, 2 Bbee, 1bbee

;

F₁ phenotypic ratio

9 Black :

3 Brown :

4 Golden

[4]

The pedigree shown in **Fig. 5** below illustrates the inheritance of coat colour in Labrador Retriever, with black coats are represent be solid symbol, those with brown coats by dotted, with golden coats by open symbols.

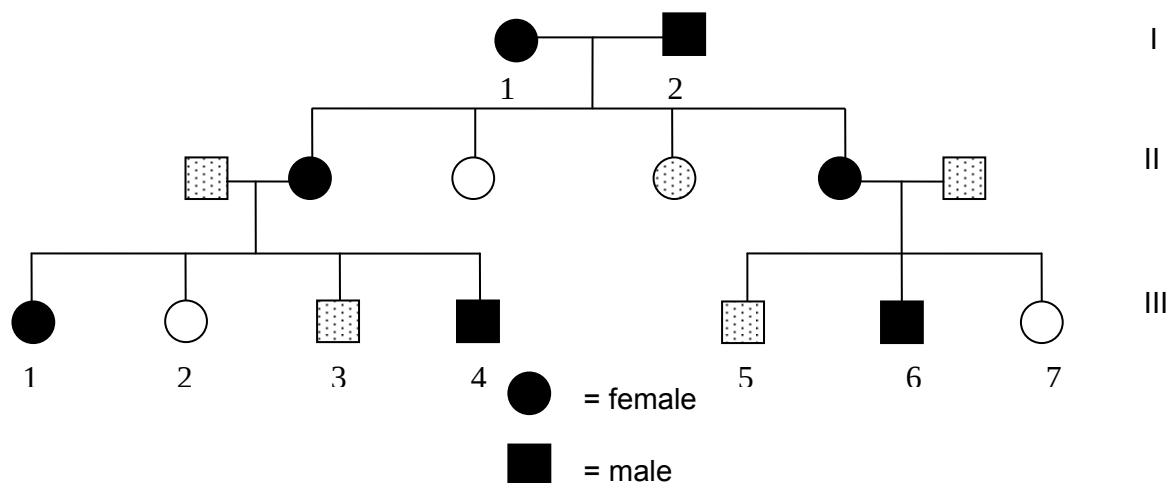


Fig. 5.1

(b) Suggest a molecular basis for the interaction described above.

epistasis;

Gene E and Gene B encode proteins/ enzymes involved in the biosynthesis of the black pigment in the dogs;

dominant allele E encode enzyme E that catalyses the synthesis of a brown/ intermediate pigment from the golden/ precursor pigment;

intermediate pigment becomes the substrate for enzyme B (encoded by dominant allele B) to synthesise the black pigment;

thus, in the absence of the E allele, the dogs are golden and in the absence of B allele, even with E present, the dogs are brown;

OR

homozygous recessive condition in the gene locus E results in the golden pigment, even if B is present;

OR

homozygous recessive condition in the gene locus E is epistatic over gene locus B;

[3]

(c) State the genotypes of the parents I-1 and I-2. Give a reason for your answer.

BbEe;

They have to be heterozygous for both genes to produce black, brown and golden phenotypes in their offspring ;

[2]

(d) State and explain the possible genotypes of III-4.

III-4: BbEe, BbEE ;

As the male parent is brown and female parent is black, the genotype has to be heterozygous for gene B/b ;

E locus can only be EE or Ee because one dominant E allele is required for the black phenotype to be expressed or ee gives a golden genotype;

[3]

[Total: 12]

- 6 **Fig. 6.1** below shows part of two nerve fibres and a synapse. The figures indicate the value in mV of the potential across the membrane between the cytoplasm of the fibres and the extracellular fluid at intervals along each fibre.

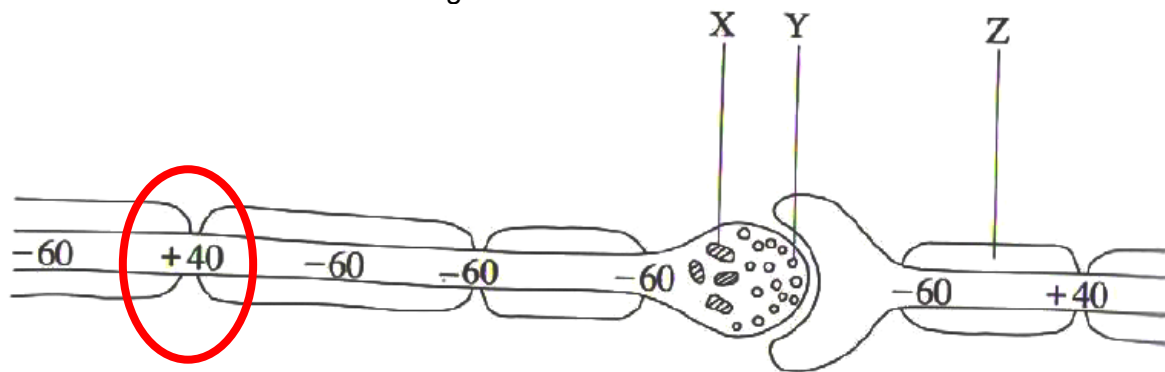


Fig. 6.1

- (a) (i) Draw a circle around one region of the diagram where an action potential exists. Explain your choice.

correct circle – 1 mark

the neuron is depolarized here with a value of +40 mV + indicating influx of Na⁺

[2]

- (ii) Explain the role of structures X and Y in the transmission of nerve impulse.

Mitochondrion (X) produces ATP via cellular respiration;

provide energy for the assembly of cytoskeleton for vesicles containing neurotransmitters to move to cell surface membrane;

provide energy to active transport of calcium ions out of cell;

Vesicles (Y) contains neurotransmitters/ acetylcholine;

vesicle membrane fuses with cell surface membrane and releases acetylcholine by exocytosis;

acetylcholine diffuses across synaptic cleft and binds to ligand-gated sodium channels, allowing diffusion of sodium ions into postsynaptic knob triggering an EPSP;

[4]

(b) **Figure 6.2a** shows a normal nerve-muscle junction.

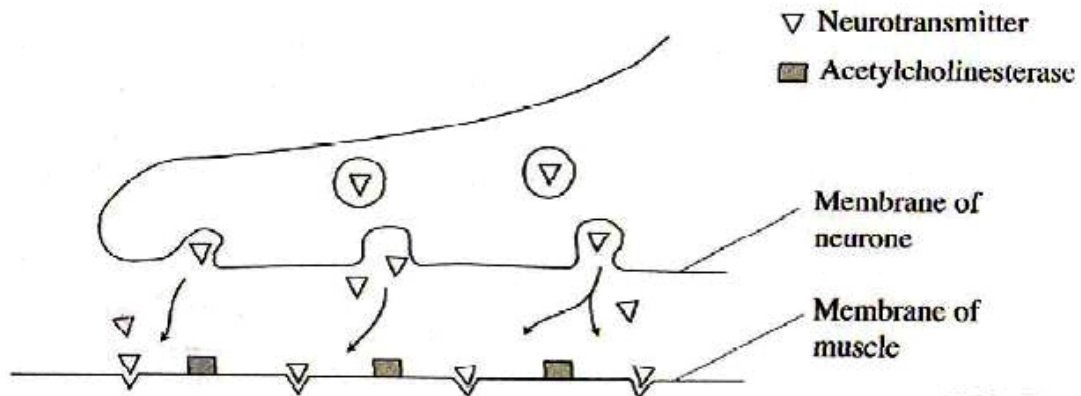


Fig. 6.2a

Using the evidence from **Fig. 6.2a** and **Fig. 6.2b**, briefly describe and explain the effect of muscle activity of (i) organophosphates and (ii) botulin toxin.

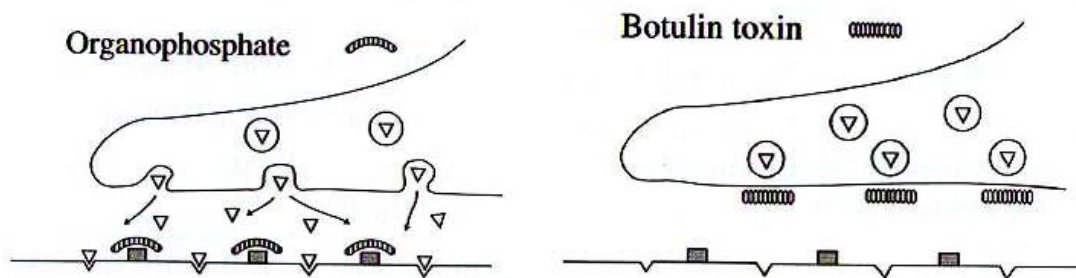


Fig. 6.2b

(i) organophosphate:

it is an inhibitor of acetylcholinesterase;
prevent enzyme from breaking down acetylcholine;
muscle activity is constantly stimulated;

[2]

(ii) botulin toxin:

prevents pre-synaptic vesicles from fusing with membrane;
thus prevent acetylcholine from being released into the synaptic cleft;
muscle cannot be stimulated;

[2]

(c) Suggest the benefit of one neurone synapsing with more than one postsynaptic neurone.

ref to dispersion of stimuli to produce a coordinated response/ communication of signals to other neurons resulting in coordination;
ref to amplification of impulse transmission: ie. 1 neuron to many neurons;
ref to same signal resulting in different outcome;

[1]

[Total: 11]

- 7 Read the passage carefully and then answer the questions that follow.



The flat periwinkles, *Littorina obtusata* and *Littorina mariae*, are two closely related marine snails that live on rocky sea shores around the coasts of the British Isles. On sheltered shores, *L. obtusata* is found mainly on the mid-shore on the seaweed *Ascophyllum nodosum* on which it feeds. *L. mariae* occurs on the lower shore on the seaweed *Fucus serratus*, feeding on the microscopic plants living on the surface of the seaweed.

The two species probably diverged from a common ancestor and they now differ markedly in both the size and colour of their shells. *L. obtusata* live for 3-4 years, breeding more than once. Their shells are larger (15-17mm) and usually green. *L. mariae* live for one year and breed only once. Their shells are smaller (9-12mm) and are usually yellow. *L. mariae* living lower on the shore is susceptible to greater predation pressure than *L. obtusata* on the mid-shore. This is due to the larger numbers of crabs which have longer foraging times on the lower shore compared to the mid-shore because of the tidal cycle.

On the lower shore, natural selection favours the annual life cycle of *L. mariae* with early maturity, rapid reproduction and short lifespan. On the mid-shore, *L. obtusata* has slow development, delayed reproduction and relatively long life-span. This species is prevented from extending its zone downwards by predations on the lower shore.

- (a) Explain why there are no *L. mariae* larger than 12mm found on the shore.

L. mariae has short life cycle/ due to genetic make-up;
larger individuals succumb to predation;
thus no large individuals can be found

[2]

- (b) Suggest why shell colour may be significant in avoiding predation.

camouflage against the different seaweed colours;
L. mariae is yellow and camouflaged against the yellow/ brown *Fucus serratus*;

[2]

- (c) Describe the role of natural selection in maintaining *L. mariae* as a separate species from *L. obtusata*.

both species are selected against when they move into the other shore position/ *L. mariae* is selected against in the mid shore and *L. obtusata* is selected against in the lower shore;
due to increased predation: for *L. obtusata* in the lower shore/ increase wave action on the larger shell size for *L. obtusata*;
decreased camouflage against the background for both species;
lack of food source;
thus there is little opportunity for the 2 species to be found in the same location and thus they do not interbreed/ no gene flow, thus remaining as separate species;

[4]

(d) Suggest a reason for the absence of

(i) *L. mariaae* from the mid-shore

lack of camouflage against the seaweed;

conditions are drier because it is higher on the inter-tidal zone, lack protection from the smaller shell size because the crabs can break through the shell;

(ii) *L. obtusata* from the lower shore

lack of camouflage; higher impact from wave action as shell is larger, thus larger surface area to volume ratio; higher predation rates;

[2]

(e) Flat periwinkles are widely distributed in all parts of the world.

Explain why different species of periwinkles found in different parts of the world can evolve similar traits.

the different species of periwinkles experience similar environment/ selection pressure/ e.g. tidal cycles, predation;
convergent evolution/ evolve independently of each other;

[2]

(f) Suggest how researchers can determine the evolutionary relationship of two periwinkle populations.

Compare DNA sequence / amino acid sequence of the same protein from both populations

The more similar the sequence, the more recent the divergence/ more recently two species branched from a common ancestor OR

The more similar the amino acid sequence, the more likely the gene coding for this protein evolved from a common gene in a shared ancestor;

nucleic acid hybridization;

[2]

[Total: 14]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8 (a) Contrast the structures and reproductive cycles of lambda phage and human immunodeficiency virus (HIV). [7]
- (b) Explain how control elements influence transcription. [6]
- (c) Describe the processes and significances, with relevant examples, of gene amplification in eukaryotes. [7]
- 9 (a) Describe the differences between continuous and discontinuous variation. [5]
- (b) Explain with examples, how embryological, anatomical and molecular homologies support Darwin's theory of evolution. [8]
- (c) Explain, with appropriate examples, how recessive alleles may be preserved in a natural population. [7]

8(a) Contrast the structure and reproductive cycles of lambda phage and human immunodeficiency virus (HIV).. [7]

Features	Lambda phage	HIV
Genome	Double-stranded linear DNA	Single-stranded positive sense RNA two copies
Capsid	Icosahedral capsid	Conical capsid
Presence of envelope	Absent	Present
Host cells	Bacteria <i>Escherichia coli</i>	T helper cells of immune system
Attachment / Adsorption	Tail fibres recognise and bind to specific receptor sites on bacterial cell wall	gp120 on viral envelope recognize and bind to specific receptor molecules (CD4+) on cell surface membrane
Entry / Penetration	Tail shaft contracts and injects the phage DNA into the cell	Virus envelope fuses with the cell surface membrane
Integration	Integration of phage DNA, as a prophage, into host cell's DNA by integrase (in lysogenic) No integration of phage DNA into host cell's DNA (in lytic)	Viral RNA is reverse transcribed to from a double stranded viral DNA by reverse transcriptase. Integration of viral DNA, as a provirus, into host cell's DNA by integrase

Synthesis of viral components	<i>(prophage is excised when phage is induced to transit from lysogenic to lytic reproductive cycle)</i> Phage DNA directs: ☐ synthesis of phage proteins ☐ replication of phage DNA in cytoplasm	Proviral genes are transcribed into RNA in nucleus RNA functions as: ☐ Viral genomes ☐ mRNAs which is translated into capsid proteins, viral proteins and glycoproteins in cytoplasm
Viral Assembly / Maturation	Phage DNA packaged inside the capsid	Capsid proteins enclose viral genome and viral proteins & assemble with glycoproteins during budding
Release	Lysozyme damages bacterial cell wall and bacterium lyses	New virus buds from the cell, surrounded by cell surface membrane embedded with viral glycoproteins

(b) Explain how control elements influence transcription. [6]

Control elements are non-coding DNA sequences that transcription factors can bind to regulate gene transcription;

Promoters found immediately upstream of start site, contain TATA box

TATA binding protein recognise and bind to TATA box in promoter and recruit general transcription factors and RNA polymerase to initiate basal level of transcription;

Enhancers are distal control elements / found further upstream from the start site which activators bind to increase rate of transcription

Via DNA loop through protein-protein interaction and stabilises the transcription initiation complex ;

Silencers are found proximal to gene / overlapping promoters which repressors bind to decreases rate of transcription;

Via impeding RNA polymerase / recruit enzymes which increases condensation of chromatin ;

(c) Describe the processes and significances, with relevant examples, of gene amplification in eukaryotes. [7]

Processes – Any 2 methods (max 4M)

1. During DNA replication, polymerase slippage occurring where DNA polymerase slip from the DNA template strand and reattach at distant site that was already replicated ;
2. The newly synthesized DNA strand contains a repeated section of DNA ;
3. Unequal crossing over of segments of non-sister chromatids of homologous chromosomes leading to gene duplication ;
4. Presence of identical repeated sequences presents a problem for precise pairing in subsequent meiosis which could lead to further unequal crossing-over events to give a string of the same gene in a row ;
5. Replicative transposition occurs by the production of an mRNA transcript of the to-be-amplified gene sequence and ;
6. reverse transcription of the mRNA to produce an additional copy of the original DNA sequence inserted into the genome ;

Significance + example (max 4M)

7. To meet the needs of the cell: The increased gene copy number can result in higher level of mRNA and polypeptide synthesis to meet the demands of the cell for the particular stage of development;
8. E.g., In *Xenopus*, the rRNA gene is amplified about one thousand times, which allows an oocyte to accumulate about 10¹² ribosomes;
9. Abnormal amplification of genes may lead to overexpression of proteins which leads to diseases;
10. E.g., Amplification of oncogenes such as *mdm2* and *myc* is associated with cancer;
11. Gene amplification may confer drug resistance;
12. E.g., Amplification of dihydrofolate reductase (DHFR) gene confers resistance to methotrexate (drug used in treating cancer);
13. Selective pressure from the environment may result in amplification of genes beneficial for organism or to confer an advantage to the organism in order to survive in a particular environment;
14. E.g., non-specific esterase genes in common house mosquito *Culex quinquefasciatus* are amplified after exposure to insecticide to confer insecticide-resistance (by binding and sequestering insecticide molecules);

9 (a) Describe the differences between continuous and discontinuous variation. [5]

Features	Continuous Variation	Discontinuous Variation
Types of classes	Characteristics fall within a range of intermediate values between 2 extremes;	Characteristics fall into distinct classes with no intermediates;
Number of genes involved	It is controlled by a number of genes/ polygenes ;	It is controlled by 1 or 2 major genes/ few genes;
Effects of genes	Each gene has a small additive effect on the character. Hence phenotypic expression is the result of combined effect of these genes/ alleles;	Effects of the genes are absolute / discernible; Or Different alleles at a single gene locus have large effects and different gene loci have quite different effects on the character;
Effects of environment;	It is greatly affected by the environment.	It is not greatly affected by the environment.
Type of inheritance;	Inheritance is quantitative.	Inheritance is qualitative.
Method of measurement;	Traits/ characteristics measured quantitatively by measuring (characters of degree).	Traits/ characteristics measured qualitatively by counting (characters of kind).
Examples;	Height/ weight	sex/ blood group

(b) Explain with examples, how embryological, anatomical and molecular homologies support Darwin's theory of evolution.. [8]

- a. Homology is similarity between different species as a result of common ancestry supporting Darwin's theory of evolution by natural selection whereby new species evolve by descent with modification from their common ancestor;
- b. In anatomical homology, comparing anatomical structures within a particular group of organisms often reveals that they are based upon a common prototype which varies with the organisms' environment;
- c. Since organisms are modified from ancestors, they retain the basic, underlying features of their ancestors, and modify these features for different functions due to natural selection;
- d. Morphological series tracks the modification to ancestral prototype through time, from one species to the next, thus supporting Darwin's concept of descent with modification;
- e. E.g. Organisms such as bats (flying), moles (burrowing), whale (swimming), retain the same, underlying basic pattern of their forelimbs, i.e. humerus, forearm and wrist with digits;
- f. In embryological homology, the more closely related two organisms are, the longer embryological development remains similar;
- g. Embryological homology shows that there is historical continuity between vertebrates as there are remnants from ancestors which persist in the embryos of descendants;
- h. E.g. All vertebrate embryos have pharyngeal pouches and these homologous throat pouches ultimately develop into structures with very different functions, such as gills in fishes and parts of the ears and throat in humans and other mammals;
- i. At the molecular level, all forms of life use the same genetic language of DNA and RNA, and the genetic code is essentially universal;
- j. Thus, likely that all species descended from common ancestors that used this code;
- k. Two species that show high molecular homology by comparing DNA / amino acid sequence probably have a common ancestor;
- l. E.g. By comparing haemoglobin amino acid sequence, humans and rhesus monkey could be closely related as there is only a difference of 8 amino acids in the haemoglobin polypeptide sequence;

(c) Explain, with appropriate examples, how recessive alleles may be preserved in a natural population. [7]

Diploidy

1. The diploid nature of most eukaryotes hides recessive alleles that are less favourable, from selection in the form of heterozygotes;
2. The latent variation is exposed to selection only when each parent carries the same recessive allele and combines the two copies in their zygote / occurs in the homozygous recessive state;
3. + any relevant e.g.;

Balanced polymorphism

4. It is the ability of natural selection to maintain stable frequencies of two or more phenotypic forms in a population;

Heterozygote advantage

5. Occurs when individuals who are heterozygous at a particular gene locus have greater survivorship and reproductive success than any type of homozygote;

6. E.g. recessive allele in the β -globin gene locus results in sickle-cell anaemia in homozygous individuals. Heterozygotes are resistant to malaria, an advantage in tropical regions. The environment favours heterozygotes over homozygous dominant individuals who are susceptible to malaria and homozygous recessive individuals who are harmed by sickle-cell anaemia.

Frequency-dependent selection

7. Occurs when the survival and reproduction of any one morph declines if that phenotypic form becomes too common in the population;
8. E.g. natural selection may favour non-poisonous butterflies that have the same colour pattern as poisonous butterflies. This system is called Batesian mimicry. It is a form of mimicry typified by a situation where a harmless species has evolved to imitate the warning signals of a harmful species directed at a common predator. When they are rare, birds will tend to avoid the mimics, because they will already have encountered a poisonous butterfly of the same appearance. But when the non-poisonous type becomes common, the previous encounters of birds with butterflies of their appearance are more likely to have been rewarding; the birds will not avoid eating them, and their fitness will be lower. The fitness of the mimics is negatively frequency-dependent.

Neutral mutations

9. Occurs because much of the DNA variation in populations probably has little or no impact on reproductive success, and thus natural selection does not affect this DNA;
10. Over time, the frequencies of alleles are not affected by natural selection but may increase or decrease as a result of genetic drift;